

* ပုံမှန်အသံသရာရှိသော ကိန်းကိန်းများ *

Continuous Random Variables

Lecture 6 RV အသံသရာရှိသော

Topics in this lecture

- Probability Density Function (PDF)
- Cumulative Distribution Function (CDF)
- Normal Distribution

RV ที่มีค่าที่ต่อเนื่องกันอยู่ใน uncountable set

Continuous RV

$X =$ RV ต่อเนื่องถ้ามี $f_X(x)$ ที่ไม่เป็นลบ

- A random variable X is called continuous if there is a non-negative function $f_X(x)$, called probability density function (PDF) of X , such that

$$P(X \in B) = \int_B f_X(x) dx$$

↳ ใช้หาค่าความน่าจะเป็น
ของ RV continuous

for every subset B of the real line

- The probability $P(a \leq X \leq b)$ can be computed from the PDF as follows

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx$$

→ prob ที่ $X = a$ เป็น 0

$$P(X = a) = ?$$

- The probability $P(X = a)$ can be computed from the PDF as follows

$$P(X = a) = \int_a^a f_X(x) dx = 0$$

↳ integrate PDF a → a

- Consequently,

Prob ที่ a จะน้อยกว่าที่ b

$$P(a \leq X \leq b) = P(a < X \leq b) = P(a \leq X < b) = P(a < X < b)$$

* ค. น. จะบ่งชี้ว่า X จะเท่ากับค่าใดค่าหนึ่ง หรือ X เป็น RV cont. = 0

$$\int_{-\infty}^{\infty} f_X(x) dx = 1$$

Some Continuous RV

RV $\hat{a}$ $\hat{b}$ a, b $\hat{a}$ $\hat{b}$ a, b $\hat{a}$ $\hat{b}$ a, b Uniform

- Uniform random variable

$$f_X(x) = \begin{cases} \frac{1}{b-a} & , x \in [a, b] \\ 0 & , \text{otherwise} \end{cases}$$

RV λ

- Exponential random variable

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & , x \geq 0 \\ 0 & , \text{otherwise} \end{cases}$$

Example 6.1

$$X = \text{expo RV} \mid \lambda = 2$$

- Let X be an exponential random variable with parameter $\lambda = 2$. Find

- $P(1 \leq X \leq 3)$ $x \geq 1 \text{ \& } x \leq 3 \text{ (1 - 3)}$

$$f_X(x) = \begin{cases} 2e^{-2x} & , x \geq 0 \\ 0 & , \text{otherwise} \end{cases} \rightarrow \text{ถ้า } x \leq 0 \text{ จะ}$$

$$\begin{aligned} P(1 \leq X \leq 3) &= \int_1^3 2e^{-2x} dx \\ &= -e^{-2x} \Big|_1^3 \\ &= e^{-2} - e^{-6} \\ &= 0.1329 \# \end{aligned}$$

จำอันที่ 6 ของ 76 นี้ให้แม่นๆ T^T

Example 6.1

- Let X be an exponential random variable with parameter $\lambda = 2$. Find
 - $P(X > 5)$

$$X = \text{expo RV} \mid \lambda = 2$$

$$f_X(x) = \begin{cases} 2e^{-2x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} P(X > 5) &= \int_5^{\infty} 2e^{-2x} dx \\ &= -e^{-2x} \Big|_5^{\infty} \\ &= \frac{1}{e^{10}} \end{aligned}$$

Expected Value from PDF

Expected Value

- The expected value of a continuous random variable X can be obtained from its PDF as follows

$$E[X] = \int_{-\infty}^{\infty} \underbrace{x f_X(x)}_{x \cdot f_X(x)} dx$$

Example 6.2

$X = \text{uniform RV on } [1, 5]$

- Let X be a uniform random variable in the interval $[1, 5]$. Find $E[X]$

$$f_X(x) = \begin{cases} 1/4, & x \in [1, 5] \\ 0, & \text{otherwise} \end{cases}$$

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$

$$E[X] = \int_1^5 \frac{1}{4} x dx$$

$$= \frac{1}{4} \left(\frac{x^2}{2} \right) \Big|_1^5$$

$$= \frac{1}{4} \left(\frac{25}{2} - \frac{1}{2} \right) = 3$$

Expected Value of $g(X)$

- The expected value of $g(X)$ can be obtained from the PDF of X as follows

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

Example 6.3

- Let X be a uniform random variable in the interval $[1, 5]$. Let $Y = 2X + 3$. Find $E[Y]$

$$f_X(x) = \begin{cases} 1/4, & x \in [1, 5] \\ 0, & \text{otherwise} \end{cases}$$

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

$$\begin{aligned} E[Y] &= \int_1^5 \frac{1}{4} (2x + 3) dx \\ &= \frac{1}{4} (x^2 + 3x) \Big|_1^5 \end{aligned}$$

$$E[Y] = \frac{1}{4} ((5^2 + 3 \times 5) - (1^2 + 3 \times 1)) = 9 \#$$

$$\begin{aligned} \text{also } E[Y] &= E[2X + 3] \\ &= 2E[X] + 3 \end{aligned}$$

$$2 \times 3 + 3 = 9 \#$$

Computing Variance

- Variance of X can be computed as follows

$$\sigma_X^2 = E[X^2] - (E[X])^2$$

where

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x) dx$$

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$

Cumulative Distribution Function (CDF)

ฟังก์ชันการแจกแจงสะสม

- A cumulative distribution function (CDF) of X is a function which describes the probability of the form $P(X \leq x)$
- If X is a continuous random variable, the CDF of X can be written as

↳ จาก PDF

$$F_X(x) = \int_{-\infty}^x f_X(t) dt$$

- If X is a discrete random variable, the CDF of X can be written as

↳ จาก PMF

$$F_X(x) = \sum_{k=-\infty}^x p_X(k)$$

Example 6.4

$X = \text{expo RV with } \lambda$

- Let X be an exponential random variable with parameter λ . Find its CDF

① PDF

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & , x \geq 0 \\ 0 & , \text{otherwise} \end{cases}$$

② CDF

$$\begin{aligned} F_X(x) &= \int_{-\infty}^x f_X(t) dt \\ &= \int_0^x \lambda e^{-\lambda t} \Big|_0^x \\ &= -e^{-\lambda t} \Big|_0^x \\ &= 1 - e^{-\lambda x} \end{aligned}$$

$$\therefore F_X(x) = \begin{cases} 1 - e^{-\lambda x} & , x \geq 0 \\ 0 & , \text{otherwise} \end{cases}$$

คุณสมบัติ

Properties of CDF

ข้อ 1 ถึง ข้อ 6

1. $F_X(x)$ is a monotonically non-decreasing function $\rightarrow$ If $x \leq y$, then $F_X(x) \leq F_X(y)$
2. $F_X(x)$ approaches 0 as x approaches $-\infty$ CDF เข้าใกล้ 0 เมื่อ x เข้าใกล้ $-\infty$
3. $F_X(x)$ approaches 1 as x approaches ∞ " 1 " ∞
4. If X is **discrete**, then is a piecewise constant function of x กราฟเป็นขั้นบันได
5. If X is **continuous**, then is a continuous function of x ฟังก์ชันไม่กระโดด (continuous)
6. If X is **discrete**, then หา PMF ของ X จากความสัมพันธ์ของ CDF ของ X ได้

$$p_X(k) = P(X \leq k) - P(X \leq k-1) = F_X(k) - F_X(k-1)$$

7. If X is **continuous**, then for those x at which the PDF is continuous

หา PDF ได้จาก
การหาฟังก์ชัน
ของ CDF

$$f_X(x) = \frac{dF_X}{dx}(x)$$

RV ที่มีการแจกแจงแบบปกติ , เกล็ด

Normal Random Variable

- A **continuous** random variable X is normally distributed if it has the **PDF of the following form**

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}, \quad -\infty < x < \infty$$

ความถี่

μ = ค่าเฉลี่ยของ X σ^2 คือการกระจายของ X
 where μ is the mean and σ^2 is the variance of X

- Standard normal, when $\mu = 0$ and $\sigma^2 = 1$

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \quad -\infty < x < \infty$$

ความถี่

Transforming Normal to Standard Normal

$\mu = 0, \text{var} = 1$
 $\Rightarrow$
 mapping RV Normal $\rightarrow$ RV StdNormal

- If X is normally distributed with mean μ and variance σ^2 , we can create a new standard normal random variable Z from X as follows

$$Z = \frac{X - \mu}{\sigma}$$

- Z is a linear transformation of a normal random variable X , so Z is also normal. The mean and variance of Z are as follow.

$$E[Z] = \frac{\overset{\mu}{\uparrow} E[X] - \mu}{\sigma} = 0$$

$$\text{VAR}[Z] = \frac{\text{VAR}[X]}{\sigma^2} = 1$$

Computing Probability from Standard Normal

- If Z is a standard normal variable, we can find $P(Z \leq z)$, defined as $\Phi(z)$, from a standard normal table

mean = 0, var = 1
 ↓
 find of z

Standard Normal Probabilities

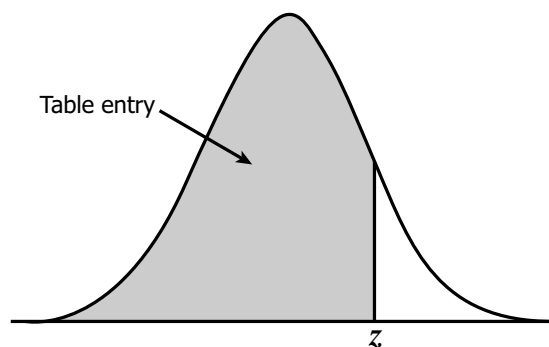


Table entry for z is the area under the standard normal curve to the left of z .

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545

$$P(Z \leq 1.35) = \Phi(1.35) = 0.9115$$

$$P(Z \leq -0.5) = 1 - \Phi(0.5)$$

$$= 1 - 0.6915$$

$$= 0.3085 \quad \#$$

ค่าที่หาได้
 ให้ 1 - ...

Probability Relating to Normal RV in General

- If X is normally distributed with mean μ and variance σ^2 , we use these steps to compute the probability $P(X \leq x)$

- ① • Create new RV, $Z = (X - \mu)/\sigma$
- ② • Find the probability $P(X \leq x)$ from the standard normal table as follows

$$\begin{aligned}
 P(X \leq x) &= P\left(\frac{X - \mu}{\sigma} \leq \frac{x - \mu}{\sigma}\right) \\
 &= P\left(Z \leq \frac{x - \mu}{\sigma}\right) \\
 &= \Phi\left(\frac{x - \mu}{\sigma}\right) \rightarrow \text{use std normal table}
 \end{aligned}$$

Example 6.5

- If X is normally distributed with mean $\mu = 2$ and variance $\sigma^2 = 4$, find the probability $P(X \leq 3)$

$$\begin{aligned}
 P(X \leq x) &= \Phi\left(\frac{x - \mu}{\sigma}\right) \\
 P(X \leq 3) &= \Phi\left(\frac{3 - 2}{2}\right) \\
 &= \Phi(0.5) \\
 &= 0.6915 \#
 \end{aligned}$$

Standard Normal Probabilities

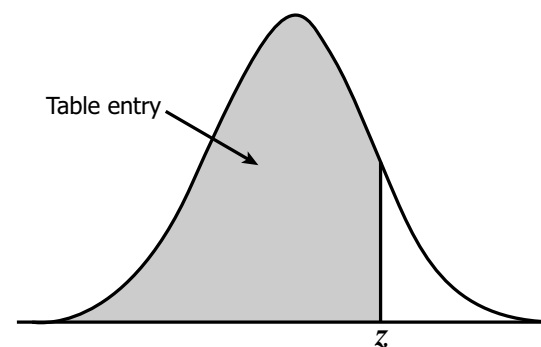


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1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545

Example 6.6

- If X is normally distributed with mean $\mu = 2$ and variance $\sigma^2 = 4$, find the probability $P(1.5 \leq X \leq 3)$

$$P(1.5 \leq X \leq 3) = P(X \leq 3) - P(X \leq 1.5)$$

$$P(X \leq 3) = \Phi(0.5) = 0.6915$$

$$P(X \leq 1.5) = \Phi\left(\frac{1.5 - 2}{2}\right)$$

$$= \Phi(-0.25)$$

$$\Phi(-0.25) = 1 - \Phi(0.25)$$

$$= 1 - 0.5987$$

$$= 0.4013$$

$$\therefore 0.6915 - 0.4013 = 0.2902 \quad \#$$

Standard Normal Probabilities

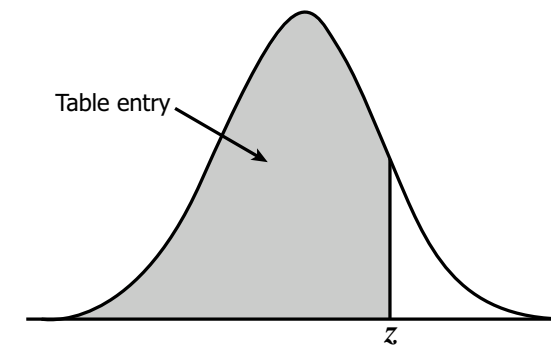


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